Isnad (إسناد) — The Trust Engine for the AI Era

An agentic, network-verified trust engine built on GSMA Open Gateway CAMARA APIs and the Nokia Network-as-Code platform.

1 · Idea Name & Submission Date

Idea Name	Isnad (إسناد) — Agentic Network Trust Engine
Submission Date	20 August 2026
Phase	Idea Phase (1 July – 23 August 2026)
Code Repository	https://github.com/AhmadShadeed32/isnad

2 · Submitter Details

Team Name	Isnad
Team Members	Ahmad Shadeed (team lead) and Yousef Al Masri
Contact Email	ahmadshadeed4321@gmail.com
Institution	Princess Sumaya University for Technology (PSUT), Amman, Jordan
Team Size	2
Country	Jordan

3 · Idea Summary

The Problem

Generative AI has broken every soft proof that someone is human and genuine. Deepfakes pass liveness and face checks, voice clones defeat call-centre verification, bots solve CAPTCHAs, and synthetic documents open accounts at scale. One-time passwords are phishable, forgeable, and add friction to every honest customer.

In MENA this lands hardest on cash on delivery, which multiple industry sources place at roughly 70–80% of regional e-commerce transactions. A COD order carries no verified buyer commitment at checkout, so fake and refused orders convert directly into wasted delivery runs, unrecoverable logistics cost, and merchant losses. At the same time, millions of creditworthy people are locked out of digital finance because they hold no formal bank or credit record.

The Proposed Solution

Isnad is an agentic trust engine. A merchant, wallet, or bank calls one API at a moment of risk — signup, checkout, payout, password reset — and asks: “Can I trust this interaction?” An investigator agent orchestrates CAMARA APIs on Nokia Network-as-Code and returns ALLOW, CHALLENGE, or DECLINE, with a full, human-readable evidence chain attached.

The one thing generative AI still cannot fake is a physical SIM card, in radio range of a physical cell tower, with years of accumulated history behind it. That truth lives inside operator networks, and CAMARA is the first time it has been programmable by third parties.

The name is literal, not decorative. Isnad (إسناد) is the classical Islamic chain of transmission — the methodology used for over a thousand years to judge whether a report could be trusted by scrutinising every link in its chain. Every Isnad verdict reproduces that structure: Human → SIM → Device → Location → History, each link backed by a specific CAMARA call. The internet never got its isnad; every verdict this engine issues is one.

Why an investigator agent, not a fixed pipeline

- **Forms a hypothesis** from the request context — for example, "new account + high value + cash on delivery" signals elevated account-takeover risk.
- **Gathers the cheapest evidence** first, beginning with the lowest-cost useful signal rather than running every API by default.
- **Escalates only when suspicion warrants it**— it calls SIM Swap, Device Swap, Location Verification, or Reachability only when its belief state justifies the cost.
- **Issues a verdict** with the full chain attached — never a black-box score.

The choice of which CAMARA API to call next is itself the agent's reasoning. This satisfies the hackathon requirement that CAMARA calls be agent-initiated decision inputs rather than user-triggered buttons, and does so more convincingly than a sequential rules pipeline.

Expected Benefits

- **Merchants and couriers:** Fewer fake and refused COD orders, and fewer wasted delivery runs, priced in pennies per check against the cost of one lost delivery.
- **Banks and wallets:** Deepfake-resistant onboarding and account-takeover defence that does not depend on OTPs, selfies, or documents.
- **Customers:** No OTP after a single consent approval; verification becomes invisible instead of adding friction.
- **Financial inclusion:** Thin-file customers with no bank record are approved on SIM tenure and location stability. Fraud tools shrink markets; this one grows one.
- **Operators:** A new, measurable revenue line for CAMARA APIs, and a repeatable reason for merchants to consume network capabilities.

4 · Project Type & GSMA Pillar Alignment

Project Type	AI agent + B2B API platform (working prototype — running code, live console, and test suite)
Primary Pillar Alignment	Anti-fraud and trusted digital identity — GSMA Open Gateway network APIs consumed as real-time trust signals rather than as user-facing features.
Secondary Alignment	Financial inclusion and digital finance enablement (Act III — approving thin-file customers on SIM tenure and location stability).
Platform	Nokia Network-as-Code (CAMARA APIs, sandbox and simulator)

Maturity	Prototype complete: /v1/verify end-to-end, signed evidence chains, live SSE agent console, 39 passing tests.

5 · Theme (One of 7)

Selected Theme	Theme 4 — Secure Fintech, Payments & Anti-Fraud Innovation
Theme APIs	The theme lists SIM Swap, Number Verification, and Device Status. Isnad uses all three as core links in the chain, and adds Location Verification, Device Reachability, Device Intelligence, and Device Roaming Status when the agent judges the extra evidence worth its cost.
Fit	Isnad is an anti-fraud layer for payments: it protects cash-on-delivery e-commerce and wallet onboarding, defends against account takeover at payout and password reset, answers vishing through Reverse Isnad, and extends payment access to customers with no formal financial record.
Adjacent Theme	Theme 1 — Trusted Digital Identity & Cross-Border Verification is the closest neighbour, and the same engine serves it. Theme 4 is selected because the wedge, the business case, and the demo all lead with payments fraud.

6 · API Usage

Seven CAMARA APIs, all available on Nokia Network-as-Code. The agent selects which to call at run time; the table shows each API's role as a link in the evidence chain.

CAMARA API	Role in the evidence chain	Agent trigger
Number Verification	Consent-based authentication at signup — proves the number belongs to this device. Replaces the OTP.	Always first: lowest-cost identity check
SIM Swap	Detects a recent SIM swap — the primary account-takeover signal.	On elevated takeover suspicion
Device Swap	Detects a new or unusual handset — the mule-phone signal.	When swap or new-account risk is live
Location Verification	Yes/no check against a claimed delivery address. Never a coordinate, never a map.	On COD and delivery-address risk
Device Reachability	Detects bot-farm and burner-phone patterns from reachability behaviour.	On synthetic-identity suspicion
Device Intelligence	Device reputation and risk scoring as a corroborating signal.	When device evidence is contested

Device Roaming Status	Detects impossible-travel patterns; doubles as a stability/income proxy for thin-file approval.	On travel anomalies and inclusion checks

Privacy design

- Location Verification returns a yes/no answer against a claim the customer already made. Isnad never receives or stores a coordinate.
- Number Verification is consent-gated: an opaque single-use state, an explicit authorization screen, then a short-lived token. Tokens are never returned in API responses and never persisted in the chain.
- No raw network signals are retained — only the resulting evidence chain, which is Ed25519-signed and independently verifiable.

7 · Team

Isnad is built by a two-person engineering team. Both members work across the stack; the areas below describe the system they built together rather than a hard split of ownership.

Member	Role
Ahmad Shadeed	Team lead · Engineering — agent and policy design, CAMARA/NaC integration, API and evidence vault
Yousef Al Masri	Engineering (co-builder) — shares the technical build across the agent, API, integration and tests

What the team built

Area	What was built
Agent & policy engine	The investigator agent, its hypothesis and belief model in log-odds, and the evidence-cost policy surface that decides which CAMARA call is worth making next.
CAMARA / NaC integration	The provider adapter for all seven CAMARA APIs on Nokia Network-as-Code, the mock simulator for stage-safe demos, and the consent-gated Number Verification flow.
API & evidence vault	POST /v1/verify with idempotent replay, Reverse Isnad, trust sessions with live revocation, and the Ed25519-signed, independently verifiable evidence chain.
Demo, tests & docs	The live SSE agent console with its four-act stage runbook, the 39-test suite, and this submission package.

Ahmad Shadeed and Yousef Al Masri · ahmadshadeed4321@gmail.com · Princess Sumaya University for Technology (PSUT), Amman, Jordan.

8 · Current Build Status

- POST /v1/verify with Idempotency-Key support; GET /v1/chains/{id} and signed-chain verification.
- POST /v1/reverse-verify — Reverse Isnad, verifying an inbound caller to the customer.

- Trust with a TTL: /v1/sessions/* keeps watching the SIM after approval and revokes a live session on mid-session swap.
- Consent-gated Number Verification flow with real NaC authorization and callback resume.
- Live agent console at /console streaming the investigation link-by-link over SSE.
- Ed25519-signed evidence vault, SQLAlchemy persistence, optional LLM planner and Redis cache, 39 passing tests.

Repository: https://github.com/AhmadShadeed32/isnad · Demo video: Isnad_Demo_Video.mp4 (2 min 44 s)